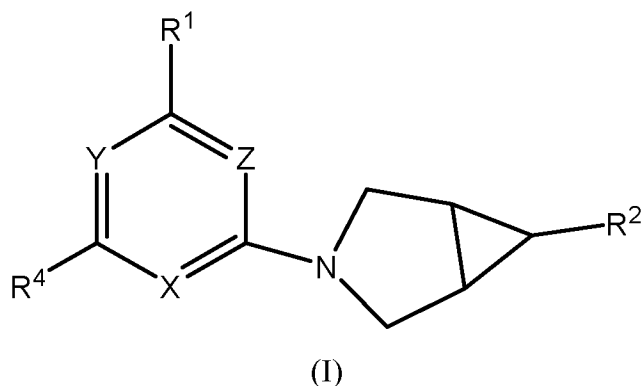


Claims

1. A compound of Formula I



or a pharmaceutically acceptable salt thereof, wherein

Y is N or C-CN;

Z is N or CH;

X is N or CR³;

provided that at least one of Y, Z, or X is N;

R¹ is C₃₋₇cycloalkyl or a 4- to 7-membered heterocyclic moiety, wherein the heterocyclic moiety contains 1 to 2 atoms independently selected from nitrogen, oxygen and sulfur, and wherein the cycloalkyl or heterocyclic moiety has 0 to 3 substituents independently selected from -C₁₋₃alkyl, and -OH, provided that there is no more than one -OH substituent; or

N(C₁₋₃alkyl)₂, NH(C₁₋₃alkyl), or NH(C₃₋₄cycloalkyl), wherein each C₁₋₃alkyl is substituted with 0 to 1 OH;

R² is -(L)_m-CON(R^N)₂, -(L)_m-SO₂R^S, -L-(CH₂)_nSO₂R^S, -L-(CH₂)_nCO₂H, -L-(CH₂)_nC(O)R^C, -L-(CH₂)_nCONHSO₂R^S, -L-(CH₂)_nSO₂NHCOR^S, -L-(CH₂)_nSO₂NHCONH₂, or -L-(CH₂)_ntetrazol-5-yl;

m is 0 or 1;

n is 0 or 1;

R^N is H or -C₁₋₃alkyl;

R^S is H or -C₁₋₃alkyl;

L is CH₂, CHF, or CF₂;

R^C is -C₁₋₄alkyloxy, -C₁₋₄alkyloxycarbonyloxy-C₁₋₄alkyloxy, or -C₁₋₄alkylcarbonyloxy-C₁₋₄alkyloxy;

R³ is H, halogen, -CN, -C₁₋₃alkyl, -OC₁₋₃alkyl, -C₁₋₃alkyl substituted with 1 to 3 halogen atoms, or -C₃₋₄cycloalkyl; and

R⁴ is cyclopropyl, cyclobutyl, or -C₁₋₃alkyl substituted with 0 to 5 halogen atoms as valency allows.

2. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein
Y is N or C-CN;
Z is N or CH;
X is CR³;
provided that at least one of Y or Z is N;
R¹ is C₃₋₇cycloalkyl or a 4- to 7-membered heterocyclic moiety, wherein the heterocyclic moiety contains 1 to 2 atoms independently selected from nitrogen, oxygen and sulfur, and wherein the cycloalkyl or heterocyclic moiety has 0 to 3 substituents independently selected from -C₁₋₃alkyl, and -OH, provided that there is no more than one -OH substituent; or
N(C₁₋₃alkyl)₂, NH(C₁₋₃alkyl), or NH(C₃₋₄cycloalkyl), wherein each C₁₋₃alkyl is substituted with 0 to 1 OH;
R² is -(L)_m-CON(R^N)₂, -(L)_m-SO₂R^S, -L-(CH₂)_nSO₂R^S, -L-(CH₂)_nCO₂H, -L-(CH₂)_nC(O)R^C, -L-(CH₂)_nCONHSO₂R^S, -L-(CH₂)_nSO₂NHCOR^S, -L-(CH₂)_nSO₂NHCONH₂, or -L-(CH₂)_ntetrazol-5-yl;
m is 0 or 1;
n is 0 or 1;
R^N is H or -C₁₋₃alkyl;
R^S is H or -C₁₋₃alkyl;
L is CH₂, CHF, or CF₂;
R^C is -C₁₋₄alkyloxy, -C₁₋₄alkyloxycarbonyloxy-C₁₋₄alkyloxy, or -C₁₋₄alkylcarbonyloxy-C₁₋₄alkyloxy;
R³ is H, halogen, -CN, -C₁₋₃alkyl, -OC₁₋₃alkyl, -C₁₋₃alkyl substituted with 1 to 3 halogen atoms, or -C₃₋₄cycloalkyl; and
R⁴ is -C₁₋₃alkyl substituted with 0 to 5 halogen atoms as valency allows.
3. The compound of claim 1 or claim 2, or a pharmaceutically acceptable salt thereof, wherein
Y is C-CN;
Z is N;
X is CR³;
R¹ is C₃₋₇cycloalkyl or a 4- to 7-membered heterocyclic moiety, wherein the heterocyclic moiety contains 1 to 2 atoms independently selected from nitrogen, oxygen and sulfur, and wherein the cycloalkyl or heterocyclic moiety has 0 to 3 substituents independently selected from -C₁₋₃alkyl, and -OH, provided that there is no more than one -OH substituent;
R² is -(L)_m-CON(R^N)₂, -(L)_m-SO₂R^S, -L-(CH₂)_nSO₂R^S, -L-(CH₂)_nCO₂H, -L-(CH₂)_nC(O)R^C, -L-(CH₂)_nCONHSO₂R^S, -L-(CH₂)_nSO₂NHCOR^S, or -L-(CH₂)_ntetrazol-5-yl;

m is 0 or 1;

n is 0 or 1;

R^N is H or -C₁₋₃alkyl;

R^S is H or -C₁₋₃alkyl;

L is CH₂, CHF, or CF₂;

R^C is -C₁₋₄alkyloxy, -C₁₋₄alkyloxycarbonyloxy-C₁₋₄alkyloxy, or -C₁₋₄alkylcarbonyloxy-C₁₋₄alkyloxy;

R³ is H, halogen, -CN, -C₁₋₃alkyl, -OC₁₋₃alkyl, -C₁₋₃alkyl substituted with 1 to 3 halogen atoms, or -C₃₋₄cycloalkyl; and

R⁴ is -C₁₋₃alkyl substituted with 0 to 5 halogen atoms as valency allows.

4. The compound of any one of claims 1 to 2, or a pharmaceutically acceptable salt thereof, wherein

Y is N;

Z is N;

X is CR³;

R¹ is C₃₋₇cycloalkyl or a 4- to 7-membered heterocyclic moiety, wherein the heterocyclic moiety contains 1 to 2 atoms independently selected from nitrogen, oxygen and sulfur, and wherein the cycloalkyl or heterocyclic moiety has 0 to 3 substituents independently selected from -C₁₋₃alkyl, and -OH, provided that there is no more than one -OH substituent;

R² is -(L)_m-CON(R^N)₂, -(L)_m-SO₂R^S, -L-(CH₂)_nSO₂R^S, -L-(CH₂)_nCO₂H, -L-(CH₂)_nC(O)R^C, -L-(CH₂)_nCONHSO₂R^S, -L-(CH₂)_nSO₂NHCOR^S, or -L-(CH₂)_ntetrazol-5-yl;

m is 0 or 1;

n is 0 or 1;

R^N is H or -C₁₋₃alkyl;

R^S is H or -C₁₋₃alkyl;

L is CH₂, CHF, or CF₂;

R^C is -C₁₋₄alkyloxy, -C₁₋₄alkyloxycarbonyloxy-C₁₋₄alkyloxy, or -C₁₋₄alkylcarbonyloxy-C₁₋₄alkyloxy;

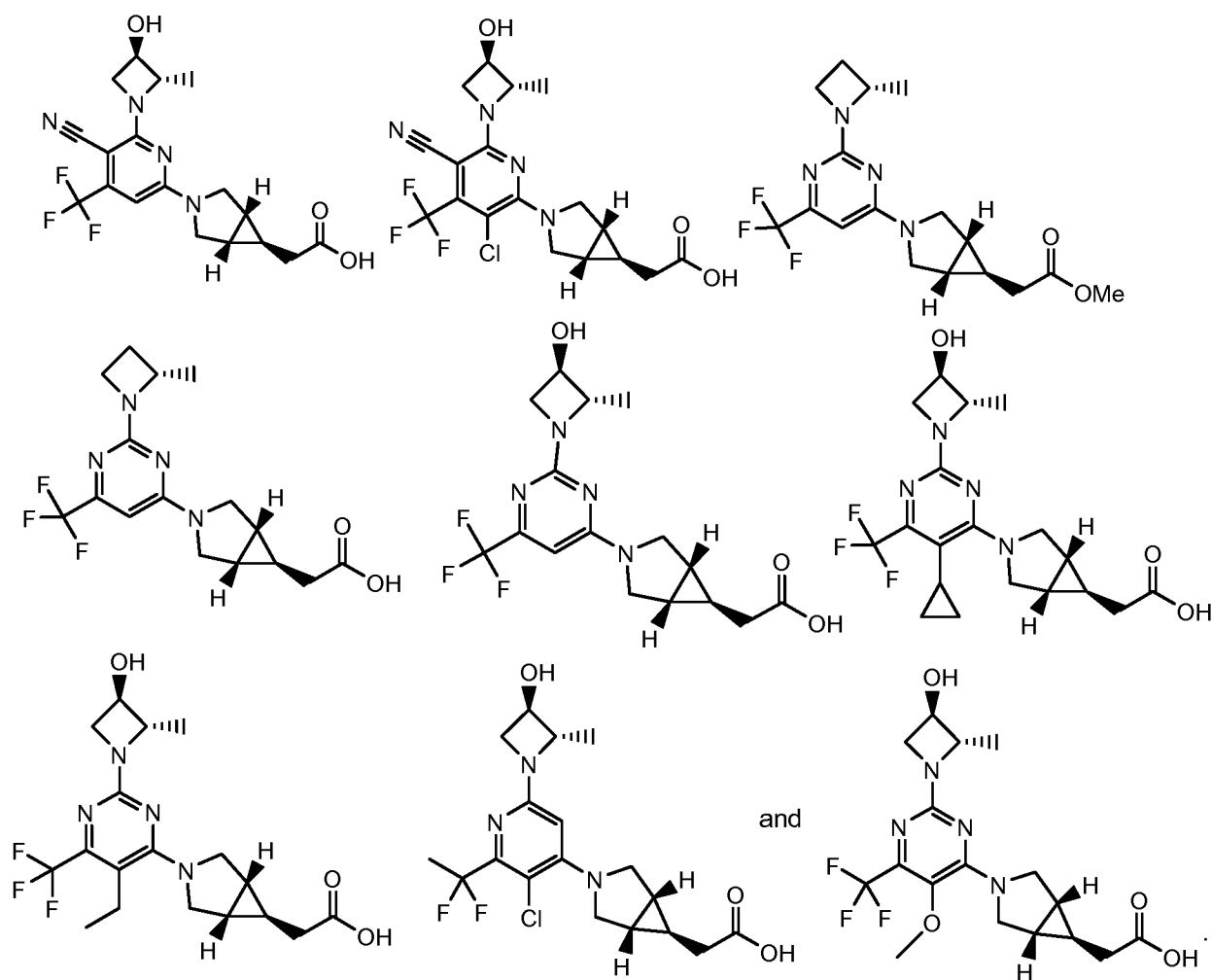
R³ is H, halogen, -CN, -C₁₋₃alkyl, -OC₁₋₃alkyl, -C₁₋₃alkyl substituted with 1 to 3 halogen atoms, or -C₃₋₄cycloalkyl; and

R⁴ is -C₁₋₃alkyl substituted with 0 to 5 halogen atoms as valency allows.

5. The compound of any one of claims 1 to 4, or a pharmaceutically acceptable salt thereof, wherein R^S is H or -CH₃.

6. The compound of any one of claims 1 to 5, or a pharmaceutically acceptable salt thereof, wherein R² is -CH₂CO₂H, -CH₂CO₂CH₃, or -CH₂CO₂CH₂CH₃.

7. The compound of any one of claims 1 to 6, or a pharmaceutically acceptable salt thereof, wherein R^3 is H, -Cl, -CH₃, -CH₂CH₃, -O-CH₃, cyclopropyl, or CN.
8. The compound of any one of claims 1 to 7, or a pharmaceutically acceptable salt thereof, wherein R^4 is -CF₃, -CHF₂, or -CF₂CH₃.
9. The compound of any one of claims 1 to 8, or a pharmaceutically acceptable salt thereof, wherein R^1 is the 4- to 7-membered heterocyclic moiety selected from azetidin-1-yl, pyrrolidin-1-yl, and piperidin-1-yl having 0 to 3 substituents independently selected from -CH₃ and -OH, provided that there is no more than one -OH substituent.
10. The compound of any one of claims 1 to 9, or a pharmaceutically acceptable salt thereof, wherein R^1 is azetidin-1-yl, having 1 to 2 -CH₃ substituents and having 0 to 1 -OH substituent, and wherein Y is C-CN and Z is N, or Y and Z are each N.
11. The compound of any one of claims 1 to 8, or a pharmaceutically acceptable salt thereof, wherein R^1 is cyclobutyl having 0 to 3 substituents independently selected from -CH₃ and -OH, provided that there is no more than one -OH substituent.
12. The compound of claim 1, or pharmaceutically acceptable salt thereof, wherein the compound is selected from



13. The compound of claim 1, or pharmaceutically acceptable salt thereof, wherein the compound is selected from

(2S,3R)-2,3-dimethyl-1-[4-[(1R,5S,6S)-6-[(methylsulfonyl)methyl]-3-azabicyclo[3.1.0]hex-3-yl]-6-(trifluoromethyl)pyrimidin-2-yl]azetidin-3-ol;

2-[(1R,5S,6R)-3-[2-[(2S)-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl]-3-azabicyclo[3.1.0]hex-6-yl]-N-(methylsulfonyl)acetamide; and

(1R,5S,6R)-3-[2-[(2S)-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl]-6-(1H-tetrazol-5-ylmethyl)-3-azabicyclo[3.1.0]hexane.

14. The compound of claim 1, or pharmaceutically acceptable salt thereof, wherein the compound is selected from

[(1R,5S,6R)-3-{5-cyano-6-[(2S,3R)-3-hydroxy-2-methylazetidin-1-yl]-4-(trifluoromethyl)pyridin-2-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid;

[(1R,5S,6R)-3-{3-chloro-5-cyano-6-[(2S,3R)-3-hydroxy-2-methylazetidin-1-yl]-4-(trifluoromethyl)pyridin-2-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid;

[(1R,5S,6R)-3-{2-[(2S)-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid;

[(1R,5S,6R)-3-{2-[(2S,3R)-3-hydroxy-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid;

[(1R,5S,6R)-3-{2-[(2S,3R)-3-hydroxy-2-methylazetidin-1-yl]-5-methyl-6-(trifluoromethyl)pyrimidin-4-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid; and

[(1R,5S,6R)-3-{5-cyano-4-(1,1-difluoroethyl)-3-fluoro-6-[(2S,3R)-3-hydroxy-2-methylazetidin-1-yl]pyridin-2-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid.

15. A compound of claim 1, or pharmaceutically acceptable salt thereof, wherein the compound is [(1R,5S,6R)-3-{2-[(2S)-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid.

16. The compound of claim 15, wherein the compound is a crystalline form of [(1R,5S,6R)-3-{2-[(2S)-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid.

17. The compound of claim 16, wherein the crystalline form is characterized substantially by the following principal powder x-ray diffraction pattern peaks expressed in terms of 2θ as measured with a copper radiation chosen from 9.0, 10.4, 15.0, and $21.4 \pm 0.2^\circ$.

18. A compound of claim 1, or pharmaceutically acceptable salt thereof, wherein the compound is [(1R,5S,6R)-3-{2-[(2S,3R)-3-hydroxy-2-methylazetidin-1-yl]-6-(trifluoromethyl)pyrimidin-4-yl}-3-azabicyclo[3.1.0]hex-6-yl]acetic acid.

19. A pharmaceutical composition comprising the compound of Formula I of any of claims 1 to 18, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient.

20. A compound of Formula I of any one of claims 1 to 18, or a pharmaceutically acceptable salt thereof, for use in a method of treating a disease, wherein the disease is selected from any one or combination of T1D, T2D, idiopathic T1D, LADA, EOD, YOAD, MODY, malnutrition-related diabetes, gestational diabetes, hyperglycemia, insulin resistance, hepatic insulin resistance, impaired glucose tolerance, diabetic neuropathy, diabetic nephropathy, kidney disease, acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, diabetic retinopathy, adipocyte dysfunction, visceral adipose deposition, obesity, eating disorders, excessive sugar craving, dyslipidemia, hyperlipidemia, hypertriglyceridemia, increased total cholesterol, high LDL cholesterol, low HDL cholesterol, hyperinsulinemia, NAFLD, steatosis, NASH, fibrosis, cirrhosis, hepatocellular carcinoma, HFI, coronary artery disease, peripheral vascular disease, hypertension, endothelial dysfunction, impaired vascular compliance, congestive heart failure, myocardial infarction, stroke, hemorrhagic stroke, ischemic stroke, pulmonary hypertension, restenosis after angioplasty, intermittent claudication, post-prandial lipemia,

metabolic acidosis, ketosis, arthritis, osteoporosis, left ventricular hypertrophy, peripheral arterial disease, macular degeneration, cataract, glomerulosclerosis, chronic renal failure, metabolic syndrome, syndrome X, premenstrual syndrome, angina pectoris, thrombosis, atherosclerosis, transient ischemic attacks, vascular restenosis, impaired glucose metabolism, conditions of impaired fasting plasma glucose, hyperuricemia, gout, erectile dysfunction, skin and connective tissue disorders, foot ulcerations, ulcerative colitis, hyper apo B lipoproteinemia, Alzheimer's Disease, schizophrenia, impaired cognition, inflammatory bowel disease, ulcerative colitis, Crohn's disease, and irritable bowel syndrome.